PM12 channel 12: integrator balance and signed ramp response

FIT hardware acceptance bench

13 November 2026

1 Setup and controls

PM12 FPGA DNA 3A17A1B402212223 was established in the previous restoration; MCU EEPROM serial 1803, the resident FPGA identity and physical PGOOD were checked for these measurements. Original INTA/INTB CPLD12 gates were restored and vendor-verified on 12 November. Today's Kasli preflight observed 683/682 rising edges on INTA/INTB. No clock profile or FPGA/CPLD image was changed.

The owner moved the MOD-DAC-28 USB cable from the direct ADC injection input J8 to the integrator stimulus input J6. CH0/CH1 are controlled through the existing SCPI entities; instrument identity FIT-CTRL-6825DD4C49E4 is checked. Every command is limited to 0–3.3 V per output. The shared local and station2 leases cover the complete hardware session. Other DAC outputs are unchanged.

CFD MCP23017: GPA7 LOW releases Integrator_PDn and enables the amplifiers; GPB5 HIGH isolates the direct-ADC injection mux. GPA3 HIGH keeps the monitor output inhibited. GPIO latches and directions are checked by readback.

The initial assumed 1.024/2.048 V input order saturated both ADCs at code 0. A mapping scan found a usable region with the reverse DAC coordinate order. The working coordinate system is therefore

$$V_{CH0} = 2.048\,\mathrm{V} + \delta/2, \qquad V_{CH1} = 1.024\,\mathrm{V} - \delta/2.$$

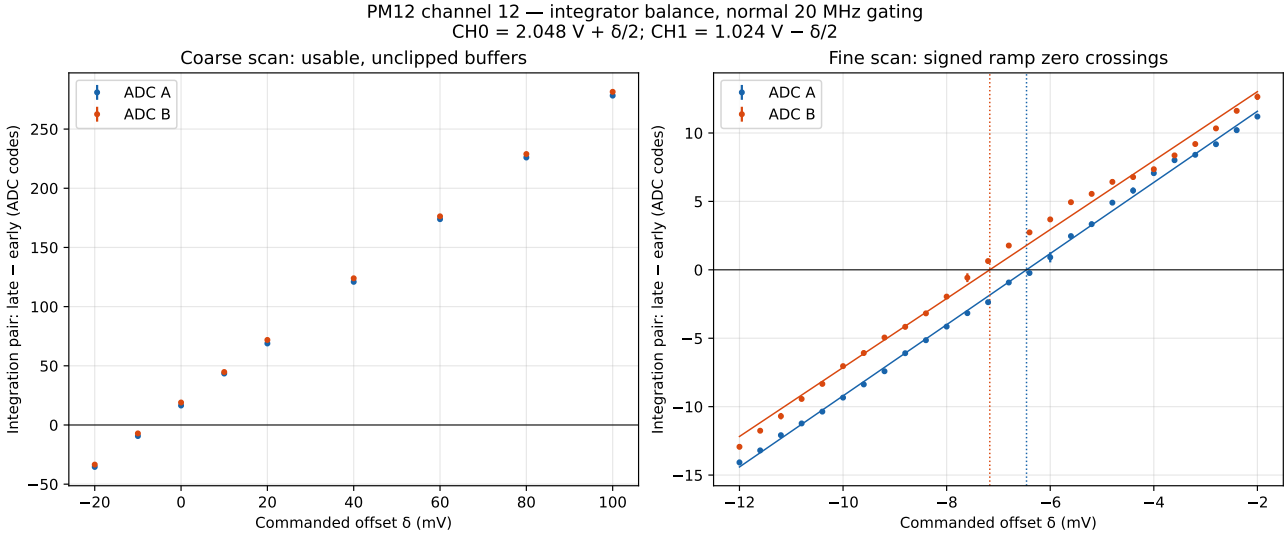
The actual differential setpoint is $1.024\,\mathrm{V} + \delta$. This is an empirically established fixture coordinate system, not an independent measurement of adapter polarity or board reference voltages.

2 Method

Integrators remain at nominal 20 MHz and ADCs at nominal 80 MHz: four ADC samples per integration/reset cycle. For each setting, frozen 4096-word FPGA buffers are downloaded over IPbus and strobe-decimated with timing/coherency validation. The coarse scan has 13 settings; the fine scan has 26 settings from -12 to -2 mV in 0.4 mV steps, with four buffers per ADC/setting.

Capture start is not tagged with integrator phase. Within each buffer, the separated four-phase template is aligned to its highest adjacent pair, assigned to integration by its strong input-dependent ramp response. The observable is late minus early ADC code, averaged over approximately 255 cycles/buffer. Clipped traces and ambiguous templates are rejected; a saturated constant output is never classified as balance. Thirty-two coarse buffers were rejected.

3 Zero crossings and response



ADC	Local zero δ (mV)	CH0 (V)	CH1 (V)	Gain (codes/V)
A	-6.319	2.044840	1.027160	2599.6
B	-7.411	2.044295	1.027705	2520.0

Zeros in the table are local interpolation of adjacent sign-changing fine-scan points. Gain is a linear fit to the full fine scan. Global fit residual RMS is 0.318/0.550 codes (A/B). Fit-only statistical uncertainties exclude DAC quantization, voltage accuracy, timing calibration and drift.

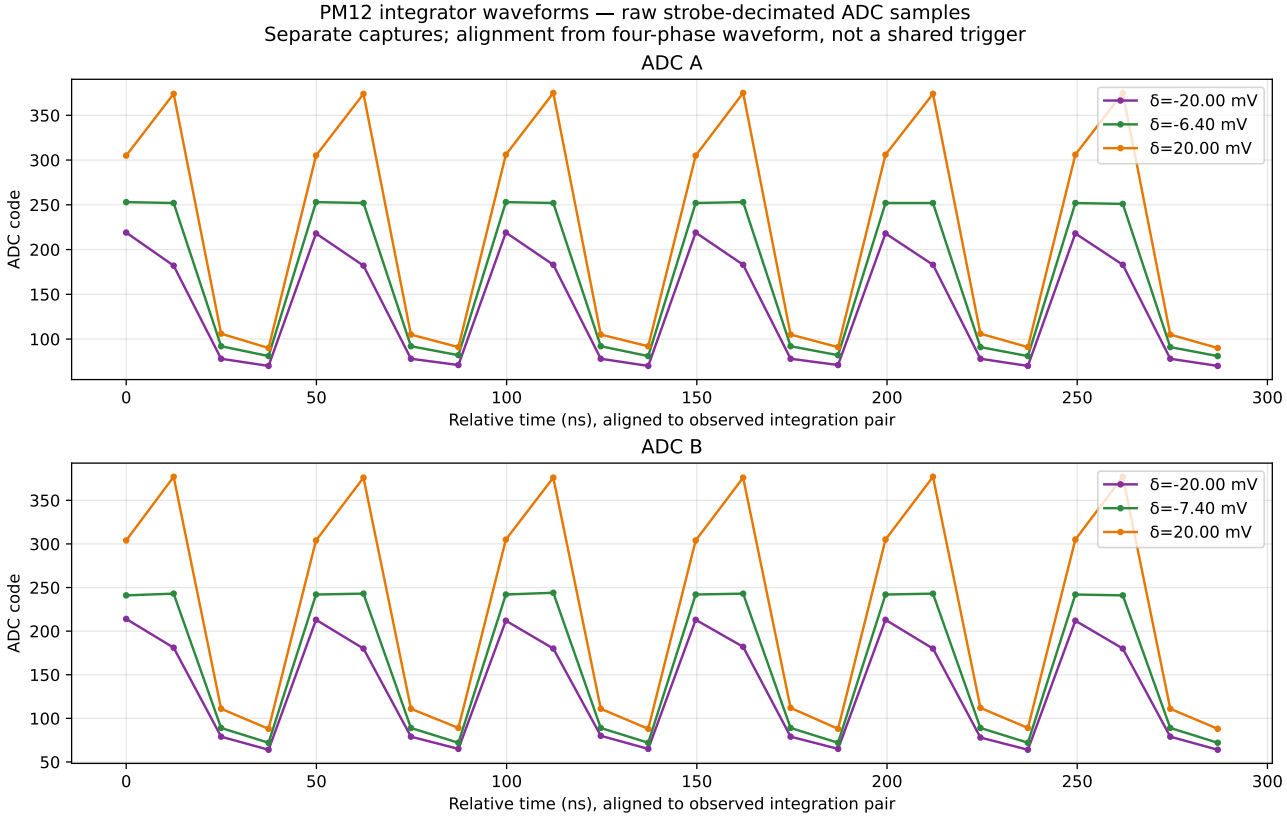
4 Independent returns to the operating points

The validation sequence revisited each operating point, measured both ADCs, and applied positive and negative 5 mV differential offsets. Each retained setting uses eight buffers, approximately 2040 cycles per ADC; the final candidate points were revisited twice. The observed near-zero changes were:

ADC	Tested CH0 (V)	Tested CH1 (V)	Change (codes/ADC interval)
A	2.0448	1.0272	-0.281
B	2.0443	1.0277	+0.329

The ±5 mV offsets produce approximately ±13 codes of change, with opposite signs. The two A/B balance points differ: one shared stimulus cannot null both exactly. A compromise setting is retained rather than declaring both perfectly flat.

5 Captured waveforms



Each trace is aligned locally, without an external trigger epoch. Timing assumes an ADC rate of 80.1574 MS/s from a 40.0787 MHz reference. No frequency counter check was made. Two adjacent samples are about 12.475 ns apart. A zero change applies to the selected integration samples; reset steps remain.

6 Final state and limits

Amplifiers remain enabled, direct ADC bypass isolated, monitor inhibited and both upstream TCA masks disconnected. DAC CH0/CH1 are parked at 2.0446/1.0274 V, between the measured A/B zero points. Other outputs are unchanged. The last capture is ADC A. No firmware, clock, EEPROM or configuration-flash write was needed.

These are whole DAC/cable/integrator/ADC-chain results against commanded voltages. They establish a signed response and an effective differential balance point, not separately measured input bias of each opamp, calibrated capacitor value, resolved intra-sample curvature, or intrinsic jitter/noise. Raw buffers, rejected probes, command receipts and analysis hashes are retained.